

DSA210-PROJECT PROPOSAL-Asude Betül Öztürk

In this project, I aim to investigate which lifestyle factors are the strongest predictors of academic success among students. Rather than only examining relationships, this study focuses on identifying the most influential variables affecting students' GPA.

The primary dataset will be obtained from the publicly available Student Lifestyle Dataset Kaggle(<https://www.kaggle.com/code/premapriya52/student-lifestyle-dataset>). This dataset includes variables such as study hours, sleep duration, social media usage, and GPA, and contains a sufficient number of observations to perform statistical analysis and machine learning.

To enrich the dataset, I will incorporate external benchmark information from reliable sources. Recommended sleep duration for young adults (7–9 hours) will be based on guidelines from the Centers for Disease Control and Prevention (https://www.cdc.gov/sleep/about_sleep/how_much_sleep.html) and the National Sleep Foundation(<https://www.sleepfoundation.org/how-sleep-works/how-much-sleep-do-we-really-need>). In addition, average screen time statistics will be referenced from Statista (<https://www.statista.com/>) and Pew Research Center(<https://www.pewresearch.org/>). Based on these sources, I will create new features such as sleep categories (low, ideal, high) and a productivity score derived from study time and screen time.

The data will be cleaned and preprocessed before analysis. Exploratory data analysis (EDA) will be conducted to examine relationships between lifestyle variables and academic performance. I will perform correlation analysis and hypothesis testing to evaluate the strength and significance of these relationships.

Furthermore, I will implement a machine learning model, such as linear regression, to predict GPA based on lifestyle factors. I will also analyze feature importance to determine which variable has the greatest impact on academic success.

The goal of this project is to provide a data-driven understanding of student productivity and identify the most critical habits that contribute to academic performance.